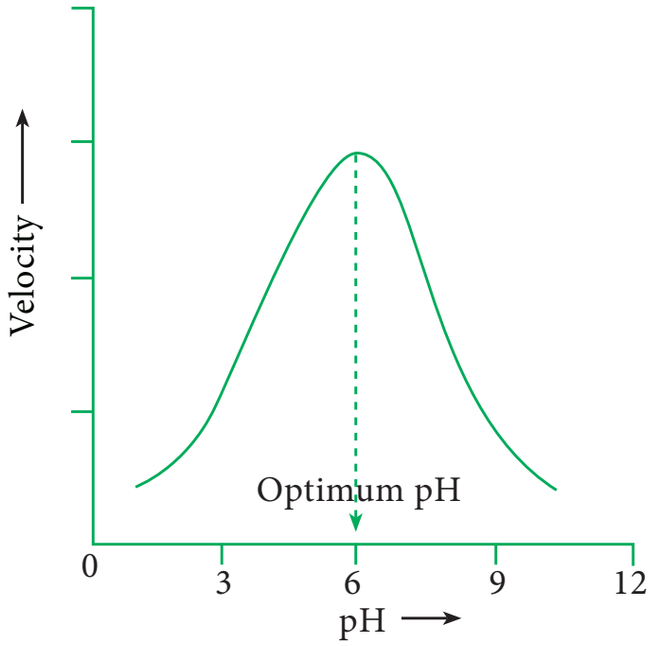


*Figure 4.3 Effect of pH*

**Note**

**Denaturation:** Nonproteolytic change in structure of a native enzyme that causes it to lose some or all of its unique or specific characteristics.

3. The optimum pH of some of the common enzymes are as follows:

| Enzyme | Optimum pH |
|---|---|
| Pepsin | 1-2 |
| Alkaline phosphatase | 10-11 |
| Acid phosphatase | 4-5 |

4. Enzymes possess only low activity or even become inactive at extreme pH values. This is due to the following reasons;

a) The hydrogen ion concentration affects the ionic charges on the active site of the enzyme.

b) Thus the extreme pH values lower the effective concentration of active form of enzyme and substrate. Therefore, the reaction velocity will be lowered.

c) Denaturation of enzyme occurs at extreme pH values.

### 4.4.2. Effect of temperature on enzyme acitivity:

1. Velocity of an enzyme reaction increases with increase in temperature upto a maximum and then declines.

2. When the velocity is plotted against temperature we obtain a plot as shown in the figure 3.4.

3. The temperature at which the enzymatic reaction has maximum velocity is known as **optimum temperature.**

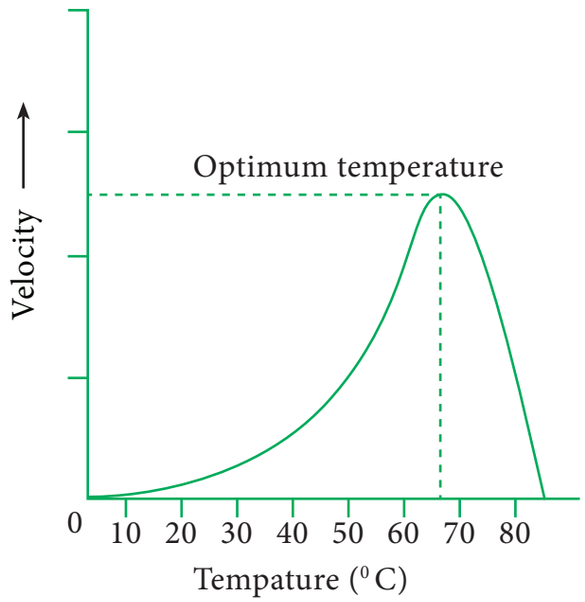


*Figure 4.4 Effect of temperature*

4. The optimum temperature of some of the common enzymes are as follows:



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